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DEPARTMENT OF COMPUTER APPLICATIONS
24MCAT213 - INTERNET OF THINGS (MCA 2025-27, SEMESTER 3)

FloraGuard IoT

Agricultural Microclimate Telemetry & Plant Pathogen Early Warning System
with SQLite3 Persistence, Containerized Backend & Real-Time Dashboard

Course Code: 24MCAT213 - Internet of Things	**Date:** November 2026
Target Hardware: ESP32 DevKit v1	**Cloud Platform:** Render.com (Docker)
Public Cloud URL: https://floraguard-iot.onrender.com	**Database:** SQLite3 (floraguard.db)

1. Abstract & Problem Statement

FloraGuard IoT is an end-to-end smart agricultural microclimate telemetry and automated plant pathogen early warning system. Using an **ESP32 microcontroller**, the system acquires real-time environmental metrics from three physical sensors: ambient air temperature and relative humidity via a digital **DHT11 sensor**, soil moisture saturation via an **analog capacitive probe**, and surface precipitation/leaf wetness via a **conductive rain sensor grid**. Telemetry readings are packaged into structured JSON payloads and transmitted over Wi-Fi via HTTP/HTTPS POST to a high-performance **FastAPI** backend service backed by an **SQLite3 relational database**. An integrated epidemiological calculation engine evaluates fungal late blight (*Phytophthora infestans*) and downy mildew threats based on the agricultural **Mills Period Infection Model**. The live readings, historical charts, and pathogen risk levels are broadcast to a responsive web dashboard via WebSockets and accessible through a visual database management interface.

2. Hardware Specifications & Interfacing Matrix

The physical node comprises four core hardware modules engineered for stable edge acquisition:

- **ESP32 Dev Module (ESP-WROOM-32):** 240MHz dual-core Xtensa LX6 processor with integrated 2.4 GHz Wi-Fi (802.11 b/g/n), 520 KB SRAM, and 12-bit ADC channels.
- **DHT11 Sensor:** Calibrated digital signal output measuring air temperature (0–50°C, ±2°C) and relative humidity (20–90% RH, ±5%).
- **Capacitive Soil Moisture Probe v1.2:** Corrosion-resistant analog probe measuring volumetric water content by dielectric permittivity.
- **Rain / Leaf Wetness Module:** Interdigital nickel-plated sensor grid tracking liquid precipitation and dew accumulation.
- **Status LED Indicator:** Active GPIO pin signaling network transmission and hardware heartbeat.

Sensor Wiring & Interfacing Pinout

Sensor Module	Module Pin	ESP32 Pin	Interface / Mode	Description
DHT11	DATA / OUT	**GPIO 4**	Digital GPIO	One-wire digital signal
Soil Moisture	AOUT	**GPIO 35**	ADC1_CH7	Analog 12-bit moisture voltage
Rain Sensor	AOUT	**GPIO 34**	ADC1_CH6	Analog 12-bit precipitation level
Status LED	Anode (+)	**GPIO 2**	Digital Output	Blinks on successful transmission
All Sensors	VCC / GND	**3.3V / GND**	Power Rails	Regulated 3.3V DC rail

3. Data Communication Architecture

The ESP32 firmware reads sensors on a 3-second cycle (3000 ms), applies a 10-sample moving average filter on analog channels to reject ADC noise, and serializes the parameters into a JSON object. The payload is posted via HTTP or HTTPS (utilizing WiFiClientSecure with certificate validation bypass for lightweight IoT transmission) to the FastAPI backend endpoint /api/telemetry:

```
{
  "device_id": "ESP32-AGRI-01",
  "temperature": 27.3,
  "humidity": 74.0,
  "soil_moisture": 44.0,
  "rain_intensity": 23.2,
  "source": "HARDWARE"
}
```

4. Database Design & Relational Persistence

To fulfill the persistent storage criteria, the application integrates an **SQLite3** database (floraguard.db) managed natively by Python. Every incoming hardware or simulated reading is validated against Pydantic schemas and permanently saved into the sensor_telemetry relational table.

Field	Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique sequential reading ID
device_id	TEXT	NOT NULL	Device identity string (e.g. ESP32-AGRI-01)
temperature	REAL	NOT NULL	Air temperature in degrees Celsius
humidity	REAL	NOT NULL	Relative humidity percentage
soil_moisture	REAL	NOT NULL	Volumetric soil moisture percentage
rain_intensity	REAL	NOT NULL	Precipitation intensity percentage
risk_score	INTEGER	NOT NULL	Mills Period pathogen index (0-100)
risk_level	TEXT	NOT NULL	Risk status (LOW / MODERATE / HIGH / CRITICAL)
diagnosis	TEXT	NOT NULL	Epidemiological pathogen diagnosis
source	TEXT	NOT NULL	HARDWARE or SIMULATION marker
timestamp	TEXT	NOT NULL	Packet time (HH:MM:SS)
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Server-side ingestion timestamp

Database Endpoints & Presentation Viewer

- **REST API (GET /api/database/records):** Supports limit parameters and source filtering (HARDWARE vs SIMULATION), returning full audit counts and JSON data.
- **Visual Database Viewer (GET /database):** A dedicated, clean web UI created specifically for viva and presentation sessions. Features real-time statistics cards, dynamic filtering, color-coded risk tags, and 10-second auto-refresh.

5. Containerization & Cloud Deployment

The application is fully containerized using **Docker** and **Docker Compose** to ensure guaranteed consistency across Windows, Linux, and Cloud environments. The image packages Python 3.11-slim, all dependencies, and persistent volume mounts for floraguard.db.
The project is publicly deployed on **Render.com** directly from the GitHub repository (https://github.com/Nehusam/floraguard-IoT), providing a high-availability cloud endpoint at **https://floraguard-iot.onrender.com**.

6. Real-Time Botanical Dashboard

- **Clean Minimalist Interface:** Built with HTML5, Tailwind CSS, Lucide icons, and Chart.js, designed specifically for rapid comprehension without visual clutter.
- **4 Live Telemetry Gauges:** Ambient Air Temperature (°C), Canopy Relative Humidity (% RH), Soil Volumetric Moisture (%), and Rain Precipitation (%).
- **Mills Period Pathogen Risk Engine:** Evaluates infection thresholds (e.g. sustained humidity >70%, temperature 18-28°C, and surface moisture) to calculate an aggregate risk score (0-100) and issue instant preventive actions.
- **WebSocket Real-Time Broadcast (/ws/telemetry):** Directly pushes live sensor events to connected browsers with sub-millisecond overhead.
- **Dual Interactive Historical Charts:** Real-time rolling graphs tracking atmospheric trends and hydrological balance over time.
- **Recent Telemetry Audit Log:** Real-time table displaying recent packets with one-click CSV export and a link to the visual database viewer.

7. Test Matrix & Experimental Verification

Test ID	Test Case	Procedure	Expected Result	Status
TC-01	Wi-Fi Connection	Power ESP32; authenticate to AP	IP assigned, serial prints success	**PASS**
TC-02	DHT11 Sampling	Read temp and humidity registers	Valid numeric °C and % values (non-NaN)	**PASS**
TC-03	Soil Probe ADC	Place probe in dry soil vs wet soil	Analog percentage scales 15% to 85%	**PASS**
TC-04	Rain Sensor Grid	Apply water droplet onto sensor plate	Rain intensity jumps from 0% to >80%	**PASS**
TC-05	HTTP/HTTPS Telemetry	ESP32 POST to /api/telemetry	HTTP 200 response with risk scoring	**PASS**
TC-06	Database Logging	Inspect SQLite3 table records	460+ rows recorded with HARDWARE tag	**PASS**
TC-07	Live Dashboard Sync	Open web dashboard on browser	Badge switches to LIVE HARDWARE DATA	**PASS**
TC-08	Cloud Deployment	Access Render public URL	Fully operational live cloud service	**PASS**

8. Conclusion

The **FloraGuard IoT** platform completely fulfills all five core assignment requirements prescribed by the MCA 24MCAT213 curriculum: (1) physical multi-sensor interfacing on ESP32, (2) reliable wireless telemetry transmission, (3) standardized containerization with Docker, (4) robust relational database persistence in SQLite3, and (5) an advanced, responsive real-time web dashboard with cloud accessibility. The solution provides a scalable, industrial-grade foundation for modern automated microclimate and precision crop disease monitoring.